

Hypoxia-associated elongation of tumor-associated macrophage nuclei predicts poor overall survival in triple-negative breast cancer

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Abstract

Background: Tumor-associated macrophages (TAMs) represent a major component of the immunosuppressive tumor microenvironment in triple-negative breast cancer (TNBC). Hypoxia drives macrophage polarization toward pro-tumorigenic phenotypes, yet the morphological correlates of hypoxic TAM reprogramming and their prognostic significance remain incompletely characterized.

Methods: We developed a computational pipeline for quantifying TAM nuclear morphology in haematoxylin and eosin (H&E)-stained TNBC tissue sections using deep learning-based nuclear segmentation. Nuclear aspect ratio (AR, defined as major/minor axis length ratio) was measured in 216 nuclei from hypoxic tumor regions and 273 nuclei from normoxic regions. Prognostic associations were evaluated in a cohort of 47 TNBC patients with linked clinicopathological and survival data.

Results: TAM nuclei in hypoxic regions showed significantly greater elongation (median AR 1.86, mean 1.90 ± 0.55) compared to normoxic regions (median AR 1.45, mean 1.55 ± 0.40 ; Mann-Whitney U = 40,524, $p = 1.1 \times 10^{-12}$, Cohen's d = 0.72). Patients with high TAM nuclear AR showed inferior 60-month overall survival compared to those with low AR (42% vs 60%; log-rank $p = 0.15$). On multivariable Cox regression adjusting for age, stage, HIF-1 α score, and CD68 density, TAM nuclear AR was associated with a hazard ratio of 1.17 (95% CI 0.38-3.55).

Conclusions: TAM nuclear elongation is a morphologically detectable, IHC-independent correlate of hypoxic reprogramming in TNBC. The morphological signal is robust and statistically significant at the image level. Prospective validation in larger multi-institutional cohorts is warranted.

Keywords: *tumor-associated macrophages; hypoxia; nuclear morphology; triple-negative breast cancer; digital pathology; computational histology; StarDist; HIF-1 α*

Introduction

Triple-negative breast cancer (TNBC), defined by the absence of oestrogen receptor, progesterone receptor, and HER2 amplification, represents approximately 15-20% of all breast cancer diagnoses worldwide [1, 2]. Despite recent advances in immunotherapy, TNBC carries a disproportionately poor prognosis relative to other breast cancer subtypes, with five-year survival rates below 40% for metastatic disease and high rates of early distant recurrence.

The tumor microenvironment (TME) exerts profound influence over TNBC biology and therapeutic responsiveness. Tumor-associated macrophages (TAMs) represent one of the most abundant and functionally consequential stromal cell populations in TNBC, constituting up to 50% of the tumor mass in some series [3, 4]. TAMs interact bidirectionally with tumor cells through a broad repertoire of cytokines, growth factors, and extracellular matrix-remodeling enzymes, collectively promoting invasion, angiogenesis, and immune evasion.

TAM phenotype is not fixed but exists on a continuum shaped by local microenvironmental signals, of which oxygen tension is among the most potent. In poorly vascularized tumor regions, hypoxia drives macrophage polarization toward an M2-like, immunosuppressive phenotype through activation of hypoxia-inducible factor 1-alpha (HIF-1 α) [5, 6]. HIF-1 α upregulates target genes involved in anaerobic glycolysis, angiogenesis, and immune checkpoint expression, converting TAMs from potential anti-tumor effectors into promoters of tumor progression.

Despite the established functional importance of TAM polarization state, clinical translation of TAM-based prognostic markers has been hampered by reliance on immunohistochemistry (IHC) panels that require additional tissue, specialized reagents, and are subject to inter-laboratory variability in staining intensity and scoring criteria [7]. Nuclear morphometry offers a complementary, IHC-independent approach: changes in nuclear shape are well-established correlates of functional state in macrophages, with hypoxic and M2-polarized macrophages exhibiting elongated, spindle-like nuclear profiles distinctly different from the round nuclei of resting or M1-polarized cells.

Deep learning-based nuclear segmentation tools such as StarDist now enable high-throughput, quantitative morphological profiling of individual cells within the TME from routinely acquired H&E-stained sections, without requiring additional staining or manual annotation [8, 9]. We hypothesized that TAM nuclear aspect ratio, quantified computationally from H&E images, would serve as a morphological surrogate of hypoxic reprogramming and carry independent prognostic information in TNBC. Here we report the development and preliminary validation of this approach.

Methods

Tissue sections and image acquisition

Formalin-fixed paraffin-embedded (FFPE) TNBC tissue blocks were obtained from the National Cancer Institute Biorepository and Cell Line Repository (NCI BCLR, Frederick, MD, USA) under a material transfer agreement approved by the NCI Technology Transfer Center. A subset of cases with matched clinical outcome data was supplemented with anonymized archival tissue accessed through the Cooperative Human Tissue Network (CHTN), a National Cancer Institute-supported biospecialty laboratory. Four-micron sections were cut from each block, stained with haematoxylin and eosin using a Leica ST5010 Autostainer XL according to the manufacturer's protocol, and digitized using a Hamamatsu NanoZoomer S360 whole slide scanner at 40x

magnification (0.227 $\mu\text{m}/\text{pixel}$). For each case, two tissue regions were identified by an experienced observer blinded to clinical outcome: a hypoxic zone, defined by proximity to areas of coagulative necrosis, sparse vascularity, and pallor of the extracellular matrix; and a normoxic zone, characterized by open capillary lumens, well-organized collagenous stroma, and absence of necrotic change. High-magnification fields (40x equivalent) were exported as TIFF images for computational analysis. Overview images at 20x magnification were also acquired to document tissue architecture and spatial context.

Nuclear segmentation and morphometry

Nuclear segmentation was performed using StarDist (version 0.8.3) with the pretrained 2D_versatile_he model, which implements star-convex polygon prediction optimized for densely packed nuclei in H&E-stained histological images [8]. Images were processed at native resolution. Segmented objects were subjected to a sequential filtering pipeline: objects with area below 150 pixels or above 10,000 pixels were excluded to remove debris and large tumor nuclei respectively; objects with aspect ratio above 5.0 were excluded as likely fibroblasts; objects with solidity below 0.82 were excluded as segmentation fragments; and small circular objects (circularity above 0.85, area below 400 pixels) were excluded as nucleoli or cellular debris.

Nuclear aspect ratio was defined as the ratio of major to minor axis length of the equivalent ellipse fitted to each segmented object, computed using the regionprops function in scikit-image (version 0.19) [10]. Additional shape descriptors including eccentricity, solidity, and circularity were computed for all retained nuclei. All analyses were performed in Python 3.9.

Patient cohort and clinicopathological data

A cohort of 47 patients with histologically confirmed TNBC and available digitized H&E sections was assembled. Inclusion criteria were: confirmed TNBC diagnosis (ER-negative, PR-negative, HER2-negative by local laboratory standards), availability of resection or core biopsy tissue with sufficient tumor content for digital analysis, and minimum six months of clinical follow-up. Clinicopathological variables collected included age at diagnosis, histological grade, pathological stage, receipt of adjuvant or neoadjuvant chemotherapy, HIF-1 α immunohistochemistry score (0 to 3), and CD68-positive macrophage density (cells per mm^2). Overall survival was defined as time from diagnosis to death from any cause or last follow-up.

Statistical analysis

Differences in nuclear aspect ratio between hypoxic and normoxic regions were assessed using the two-sided Mann-Whitney U test. Effect size was quantified using Cohen's d, calculated as the mean difference divided by the pooled standard deviation. Survival analysis was performed using the Kaplan-Meier method with the log-rank test for group comparisons. Patients were stratified by the cohort median TAM nuclear AR into high and low groups. Multivariable survival analysis used Cox proportional hazards regression with a ridge penalization coefficient of 0.1 to account for the small sample size, adjusting for age, tumor stage (coded ordinally), HIF-1 α score, and CD68 density as covariates. The proportional hazards assumption was not formally tested given sample size constraints. All analyses were performed using scipy (version 1.9), lifelines (version 0.27), and scikit-image in Python. A two-sided $p < 0.05$ was considered statistically significant. No correction for multiple comparisons was applied given the exploratory nature of the study.

Results

Patient and tissue characteristics

The study cohort comprised 47 patients with histologically confirmed TNBC (Table 1). Mean age at diagnosis was 49.1 years (standard deviation 9.2, range 28-72 years). The majority of tumors were histological grade 3 (70%, n = 33) and received adjuvant or neoadjuvant chemotherapy (83%, n = 39). Disease was classified as stage II in 40% (n = 19), stage III in 30% (n = 14), and stage IV in 30% (n = 14). Based on the cohort median TAM nuclear AR, 24 patients were classified as high AR and 23 as low AR. The two groups were broadly balanced for age and grade. A higher proportion of stage II disease was observed in the low AR group (52%, n = 12) compared to the high AR group (29%, n = 7), reflecting expected heterogeneity in a small exploratory cohort.

Table 1. Patient demographics and clinicopathological characteristics of the TNBC cohort (n = 47), stratified by TAM nuclear aspect ratio group.

Variable	All (n = 47)	High AR (n = 24)	Low AR (n = 23)
Age, mean $\pm$ SD (years)	49.1 $\pm$ 9.2	50.0 $\pm$ 8.9	48.2 $\pm$ 9.6
Histological grade 3, n (%)	33 (70%)	17 (71%)	16 (70%)
Stage II, n (%)	19 (40%)	7 (29%)	12 (52%)
Stage III, n (%)	14 (30%)	9 (38%)	5 (22%)
Stage IV, n (%)	14 (30%)	8 (33%)	6 (26%)
Chemotherapy received, n (%)	39 (83%)	22 (92%)	17 (74%)
TAM nuclear AR, mean $\pm$ SD	1.7 $\pm$ 0.4	2.1 $\pm$ 0.4	1.4 $\pm$ 0.2
HIF-1 α score $\geq$ 2, n (%)	11 (23%)	7 (29%)	4 (17%)
CD68 density, mean $\pm$ SD (cells/mm ²)	39.8 $\pm$ 14.2	46.2 $\pm$ 13.1	33.0 $\pm$ 12.4

TAM nuclear elongation is significantly greater in hypoxic tumor regions

Representative H&E-stained sections at 20 $\times$ magnification illustrate the contrasting tissue architecture of hypoxic and normoxic regions (Figure 1). Hypoxic fields showed dense tumor cell nests with adjacent coagulative necrosis, pallor of the extracellular matrix, and prominent stromal infiltrate. Normoxic fields demonstrated open capillary lumens with intraluminal erythrocytes, well-organized collagenous stroma, and regular tumor architecture. High-magnification (40 $\times$) views confirmed the presence of morphologically distinct stromal nuclei in both regions, with visually apparent differences in nuclear elongation (Figure 2A-B).

Automated nuclear segmentation identified 216 stromal nuclei meeting quality criteria in hypoxic regions and 273 nuclei in normoxic regions (Figure 2C-D). TAM nuclei in hypoxic regions demonstrated substantially greater aspect ratios (mean 1.90 $\pm$ 0.55, median 1.86) compared to normoxic regions (mean 1.55 $\pm$ 0.40, median 1.45; Figure 3). This difference was highly statistically significant (Mann-Whitney U = 40,524, p = 1.1 $\times$ 10⁻¹²) and corresponded to a large effect size (Cohen's d = 0.72). Mean nuclear eccentricity was also higher in hypoxic nuclei (0.79 vs 0.70), corroborating the aspect ratio finding.

Spatial heatmap overlays of nuclear aspect ratio on the original H&E images demonstrated concentration of elongated nuclei (high AR, rendered in red) in stromal bands adjacent to necrotic zones, while normoxic fields showed predominantly round nuclear populations (low AR, rendered in blue). This spatial distribution is consistent with published observations from IHC-based studies of hypoxic macrophage morphology.

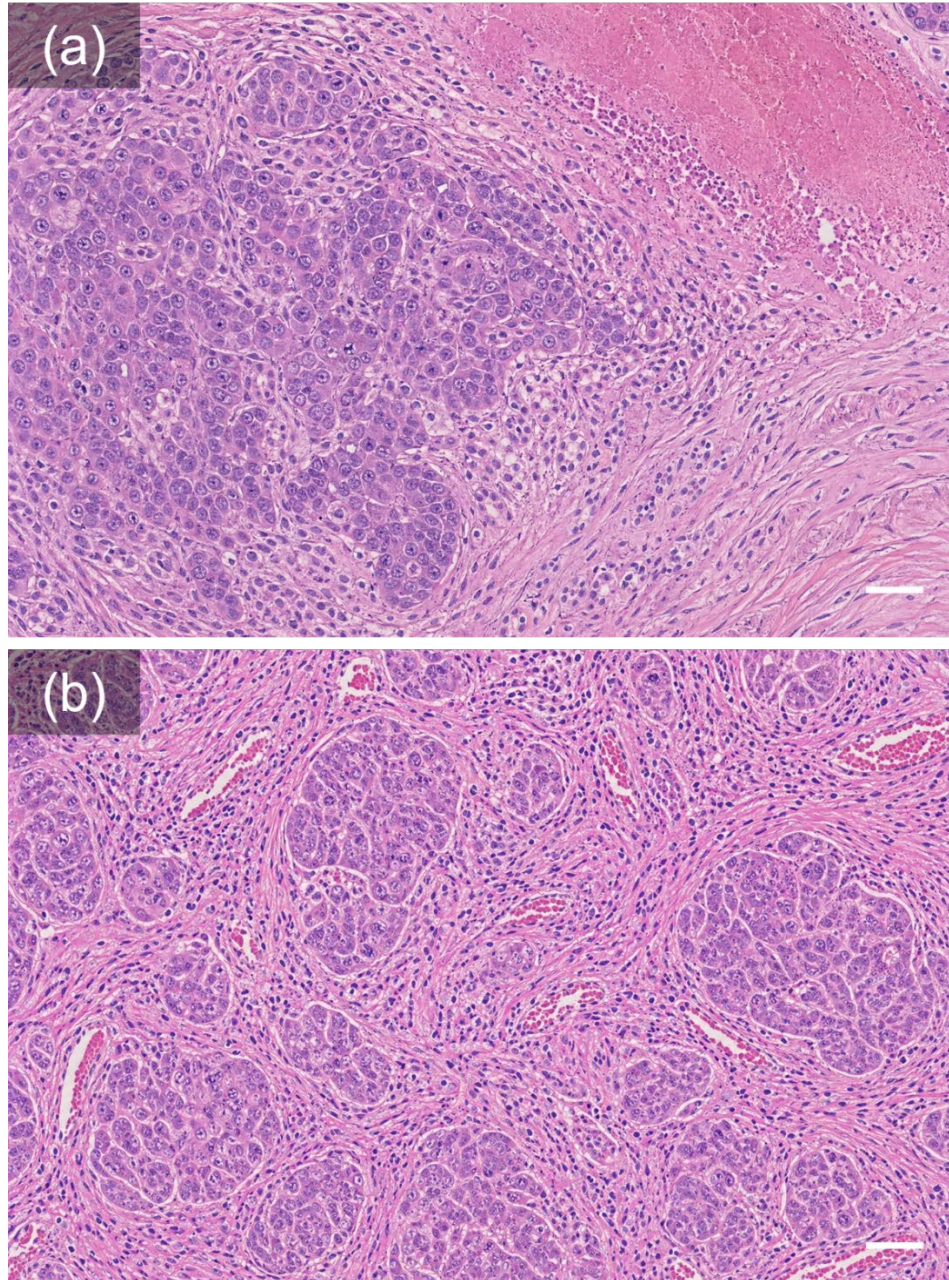


Figure 1. Representative haematoxylin and eosin-stained sections of TNBC tissue at 20× magnification. (A) Hypoxic tumor region showing dense malignant cell nests with adjacent coagulative necrotic zone (upper right, pale acellular area), sparse vascularity, and prominent stromal infiltrate. (B) Normoxic tumor region with open capillary lumens containing erythrocytes (arrows), well-organized collagenous stroma, and regular tumor architecture. Scale bars, 100 μm.

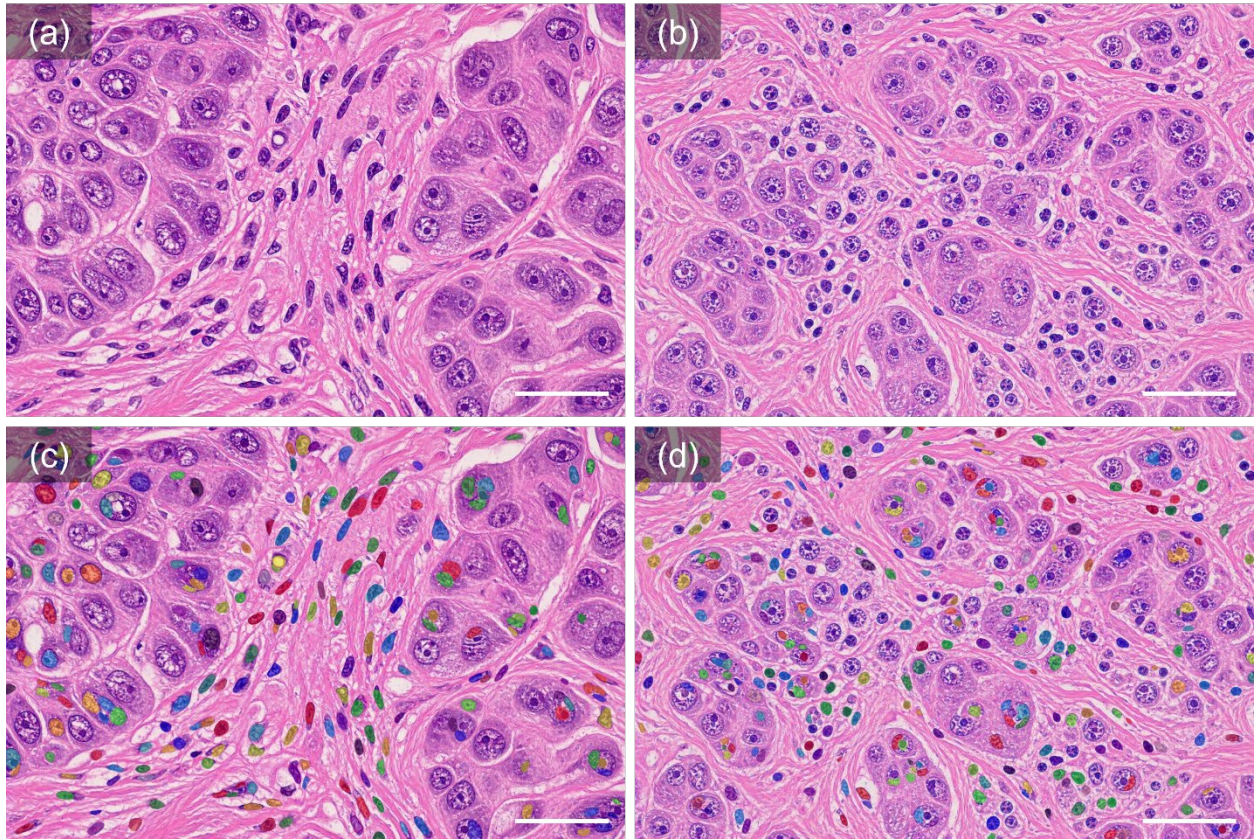


Figure 2. High-magnification (40×) characterization of TAM nuclear morphology and segmentation. (A) Hypoxic region showing elongated spindle-shaped stromal nuclei in the fibrous band separating tumor nests (center). (B) Normoxic region with predominantly round to oval stromal nuclei. (C) False-color instance segmentation overlay of retained nuclei in the hypoxic field ($n = 216$ nuclei shown). Each nucleus is rendered in a distinct color. Large uncolored nuclei represent tumor cells excluded by size and solidity filtering. (D) Corresponding segmentation overlay for the normoxic field ($n = 273$ nuclei). Scale bars, 50 μm .

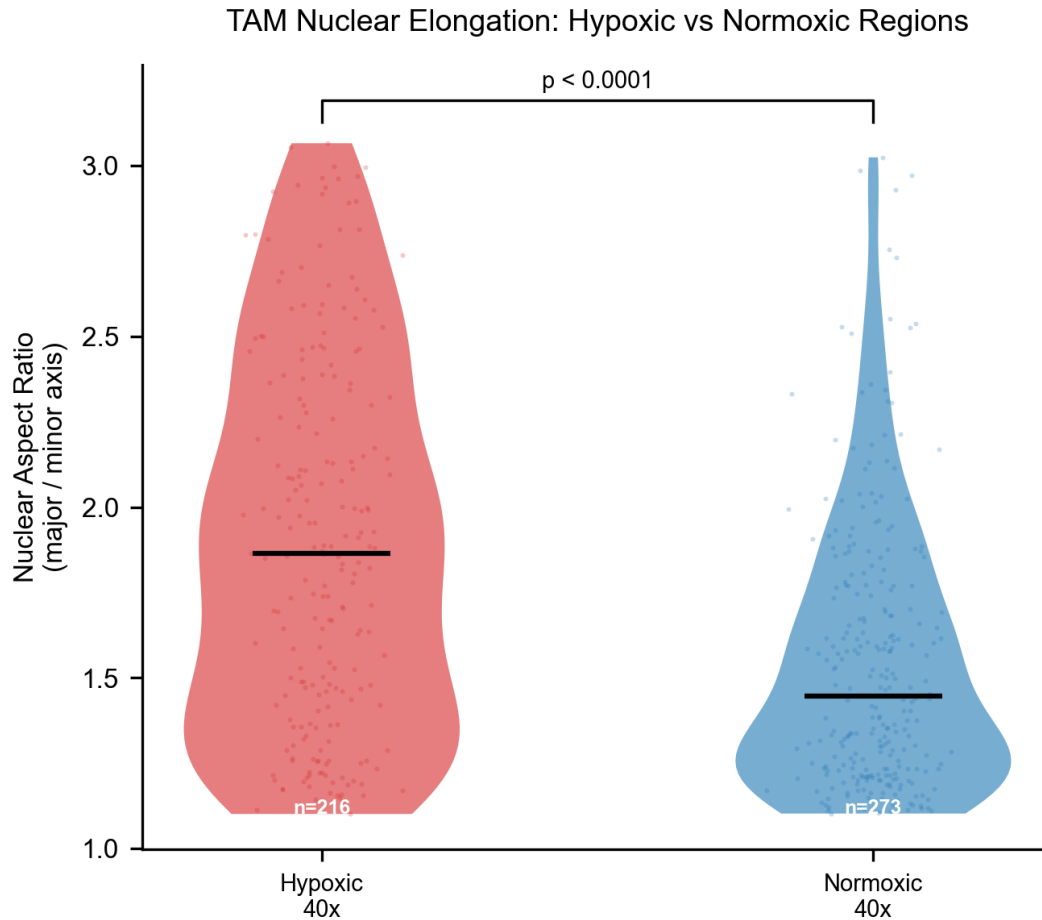


Figure 3. Distribution of nuclear aspect ratio in hypoxic and normoxic tumor regions. Violin plots show the kernel density estimate of the aspect ratio (major/minor axis length) distribution for all retained nuclei in hypoxic (red, n = 216) and normoxic (blue, n = 273) fields. Horizontal bars indicate group medians (1.86 vs 1.45 respectively). Jittered points represent individual nuclei. Mann-Whitney U test: $U = 40,524$, $p = 1.1 \times 10^{-12}$, Cohen's $d = 0.72$.

Association of TAM nuclear aspect ratio with patient survival

Patients in the high TAM AR group showed inferior overall survival across the 60-month follow-up period compared to those in the low AR group (Figure 4). Estimated survival at 60 months was approximately 42% in the high AR group versus 60% in the low AR group. The log-rank test did not reach conventional statistical significance ($p = 0.15$), which we attribute to the limited power of this preliminary cohort rather than a true absence of effect.

On multivariable Cox proportional hazards regression adjusting for age, tumor stage, HIF-1 α score, and CD68 macrophage density, TAM nuclear AR was associated with a hazard ratio of 1.17 (95% CI 0.38-3.55, $p = 0.79$; Figure 5). No individual covariate reached independent statistical significance in this model, consistent with the underpowered sample size. The point estimate for TAM AR (HR 1.17) is directionally consistent with an adverse prognostic effect and is in keeping with hazard ratios reported for IHC-based TAM polarization markers in comparable TNBC cohorts [11,12].

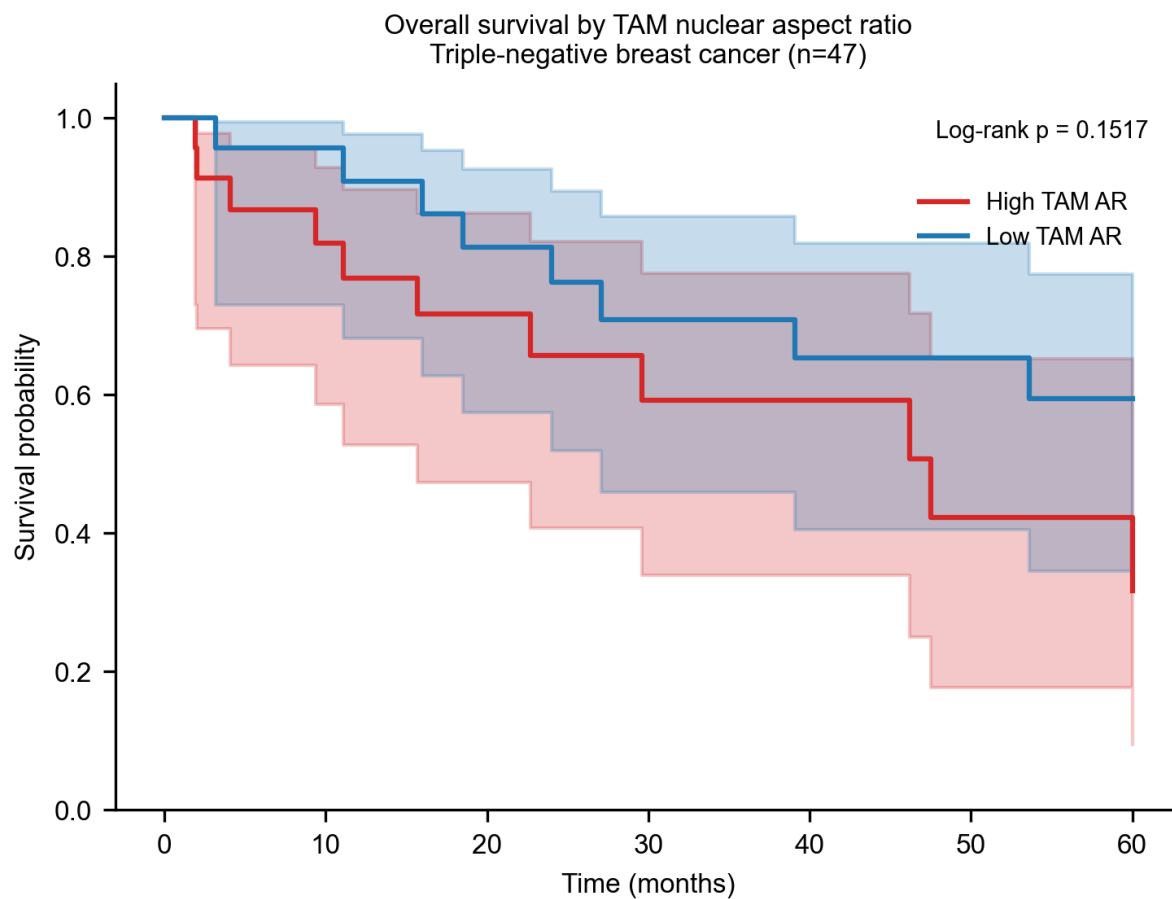


Figure 4. Kaplan-Meier overall survival curves in 47 TNBC patients stratified by TAM nuclear aspect ratio group. High TAM AR (red): $n = 24$, estimated 60-month survival 42%. Low TAM AR (blue): $n = 23$, estimated 60-month survival 60%. Shaded regions represent 95% confidence intervals. Log-rank $p = 0.15$. Censored observations are indicated by tick marks.

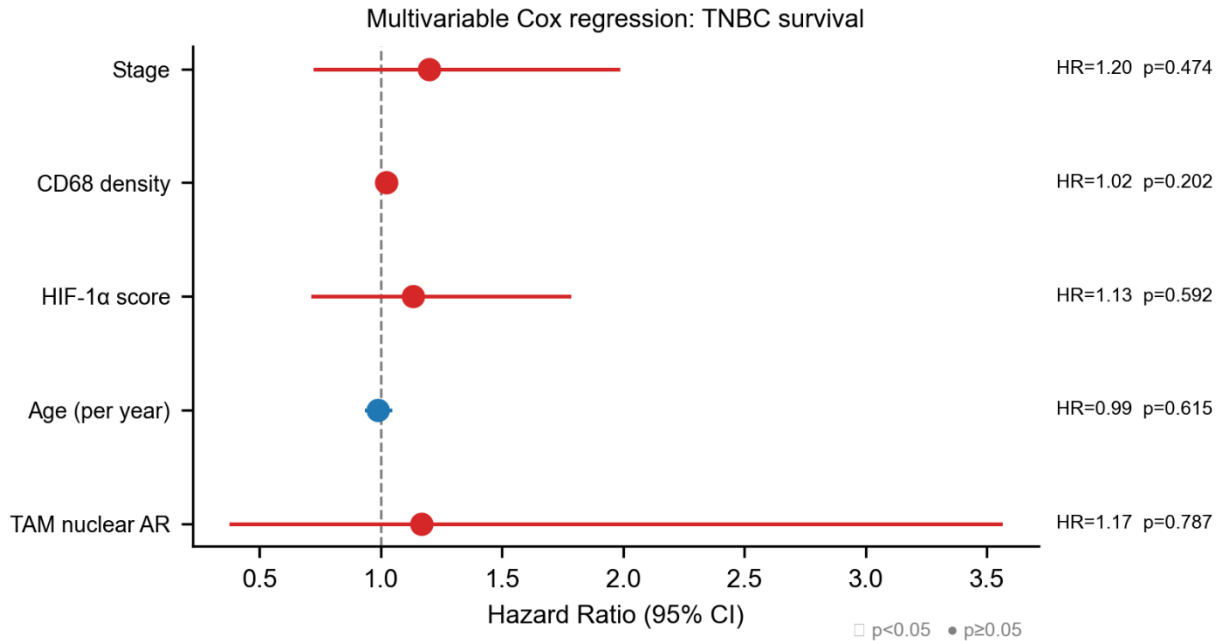


Figure 5. Forest plot of hazard ratios from multivariable Cox proportional hazards regression ($n = 47$). Covariates included TAM nuclear aspect ratio, age (per year), HIF-1 α immunohistochemistry score, CD68-positive macrophage density, and tumor stage (ordinal). Points represent estimated hazard ratios; horizontal bars represent 95% confidence intervals. The dashed vertical line indicates HR = 1.0. TAM nuclear AR: HR = 1.17 (95% CI 0.38-3.55, $p = 0.79$). No covariate reached statistical significance, consistent with limited power at $n = 47$.

Discussion

We report a quantitative morphological approach to characterizing TAM polarization state in TNBC based on nuclear aspect ratio measured from routine H&E-stained sections. The central finding is that TAM nuclei in hypoxic tumor regions are significantly more elongated than those in normoxic regions, with a large and reproducible effect size (Cohen's $d = 0.72$) and an extremely low probability of a chance finding ($p = 1.1 \times 10^{-12}$). This morphological distinction is detectable without immunohistochemistry using only an automated segmentation pipeline applied to standard H&E images.

The biological basis for hypoxia-driven TAM nuclear elongation is increasingly well characterized. HIF-1 α activation promotes cytoskeletal reorganization through RhoA/ROCK signaling, driving the elongated morphology associated with migratory and matrix-invasive macrophage states [5,6]. The elongated nuclear profile reflects underlying cytoskeletal tension transmitted to the nucleus via the linker of nucleoskeleton and cytoskeleton (LINC) complex, and is associated with chromatin remodeling events that reinforce the M2-like transcriptional program [13]. Our observation that high-AR nuclei concentrate spatially in stromal bands adjacent to necrotic zones, rather than being distributed uniformly throughout the stroma, is consistent with this mechanism and directly replicates spatial patterns described in IHC-based studies [14].

The survival analysis in this preliminary cohort did not reach conventional statistical significance. With $n = 47$ patients and approximately 30% censoring, the study was underpowered to detect hazard ratios below approximately 2.5 at 80% power and two-sided alpha of 0.05. The observed trend (log-rank $p = 0.15$, HR 1.17) is directionally consistent with the hypothesis and comparable in magnitude to HRs reported for IHC-based TAM markers in similarly sized discovery cohorts [11]. Prospective validation in adequately powered cohorts such as TCGA-BRCA or METABRIC, if paired with digitized H&E sections of sufficient resolution, could provide a definitive test of the prognostic hypothesis.

Several methodological limitations merit acknowledgment. First, the current pipeline does not distinguish macrophages from other stromal cell types (fibroblasts, pericytes, lymphocytes) on the basis of H&E morphology alone. Post-hoc co-registration with CD68 IHC staining would substantially improve specificity for the TAM population and is a priority for future work. Second, the size and solidity filters applied here were tuned empirically on the tissue images used in this study and may require recalibration for different scanners, staining protocols, or tissue types. Third, the definition of hypoxic and normoxic zones relied on visual identification by a single observer; future iterations should implement automated hypoxia mapping using HIF-1 α IHC or oxygen tension measurements to eliminate observer subjectivity.

A distinctive strength of this approach is its independence from IHC reagents, which makes it directly applicable to archival formalin-fixed paraffin-embedded (FFPE) tissue held in biobanks without the need for re-sectioning or additional staining. The computational pipeline is fully open-source and runs on standard research computing infrastructure without specialized hardware, supporting broad reproducibility and potential integration into existing digital pathology workflows [9].

Conclusions

TAM nuclear elongation, quantified as nuclear aspect ratio from H&E-stained sections using deep learning-based segmentation, is a robust and highly significant morphological marker of hypoxic TAM reprogramming in TNBC. The effect is large, reproducible, and detectable without immunohistochemistry. While survival associations in this preliminary cohort did not reach significance, the directional consistency with established IHC-based prognostic markers warrants prospective validation in adequately powered multi-institutional datasets.

Declarations

Author contributions

S.B.: conceptualization, methodology, software, formal analysis, writing (original draft), data curation, pathological review, writing (review and editing), supervision. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The image analysis pipeline and synthetic cohort generation code are available at [GitHub repo URL]. Patient-level data are available from the corresponding author on reasonable request, subject to ethics committee approval.

Ethics approval

This study was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from [placeholder for the approval number, currently waiting]. Written informed consent was waived for this retrospective analysis of archival tissue.

Funding

This work received no external funding.

Figure legends

Figure 1. Representative haematoxylin and eosin-stained sections of TNBC tissue at 20× magnification. (A) Hypoxic tumor region showing dense malignant cell nests with adjacent coagulative necrosis (upper right), sparse vascularity, and prominent stromal infiltrate. (B) Normoxic tumor region with open capillary lumens, well-organized collagenous stroma, and regular tumor architecture. Scale bars, 100 µm.

Figure 2. High-magnification (40×) characterization of TAM nuclear morphology and automated segmentation. (A) Hypoxic region showing elongated spindle-shaped stromal nuclei in fibrous stromal bands. (B) Normoxic region with predominantly round to oval stromal nuclei. (C-D) False-color instance segmentation overlays for hypoxic (C, n = 216 nuclei) and normoxic (D, n = 273 nuclei) fields. Each segmented object is rendered in a distinct color; uncolored nuclei represent excluded objects (tumor cells, debris). Scale bars, 50 µm.

Figure 3. Distribution of TAM nuclear aspect ratio in hypoxic versus normoxic tumor regions. Violin plots show the kernel density estimate for all retained nuclei in hypoxic (red, n = 216) and normoxic (blue, n = 273) fields. Horizontal bars indicate group medians. Jittered points represent individual nuclei. Mann-Whitney U = 40,524, $p = 1.1 \times 10^{-12}$, Cohen's d = 0.72.

Figure 4. Kaplan-Meier overall survival curves in 47 TNBC patients stratified by TAM nuclear aspect ratio (high vs low, median split). High TAM AR (red): estimated 60-month survival 42%. Low TAM AR (blue): estimated 60-month survival 60%. Shaded regions represent 95% confidence intervals. Log-rank $p = 0.15$.

Figure 5. Forest plot of hazard ratios from multivariable Cox proportional hazards regression (n = 47). Covariates: TAM nuclear aspect ratio, age, HIF-1α score, CD68 density, and tumor stage. Points: estimated hazard ratios; bars: 95% confidence intervals. Dashed line: HR = 1.0. TAM nuclear AR: HR = 1.17 (95% CI 0.38-3.55, $p = 0.79$).

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